DC Power Flow

DC power flow is a linearized approximation of AC power flow, suitable for economic dispatch an
d contingency analysis.

DC Approximations

1. **Lossless lines** - Ignore resistance ($R \approx 0$)
2. **Flat voltage** - All bus voltages = 1.0 pu
3. **Small angles** - $\sin(\theta) \approx \theta$, $\cos(\theta) \approx 1$

Result: Power flow depends only on bus angles (θ) and line reactances (X).

Bus Number Mapping

Power system bus numbers may not be contiguous (e.g., case300 has non-sequential bus IDs). Alwa
ys create a mapping from bus numbers to 0-indexed array positions:

```python
# Create mapping: bus_number -> 0-indexed position
bus_num_to_idx = {int(buses[i, 0]): i for i in range(n_bus)}

# Use mapping for branch endpoints
f = bus_num_to_idx[int(br[0])]  # NOT br[0] - 1
t = bus_num_to_idx[int(br[1])]
```

Susceptance Matrix (B)

Build from branch reactances using bus number mapping:

```python
# Run: scripts/build_b_matrix.py
# Or inline:
bus_num_to_idx = {int(buses[i, 0]): i for i in range(n_bus)}
B = np.zeros((n_bus, n_bus))
```